

Fraud Detection Analysis & Risk Interpretation

An Exploratory and Machine-Learning-Oriented Analysis of Credit Card Transactions

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Project Type	Major Project
Domain	Data Science / Financial Risk Analytics
Primary Dataset	Credit Card Fraud Detection Dataset (MLG-ULB)
Tools	Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
Project Period	Data Science Internship

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Acknowledgement

I would like to express my sincere gratitude to everyone who supported me during the completion of this major project. This project provided an opportunity to apply data science concepts to a realistic financial-risk problem and to understand how exploratory analysis, data preparation and machine learning can contribute to fraud detection.

I am thankful to the internship program and its learning resources for providing the project framework and practical direction. I also acknowledge the Machine Learning Group at Université Libre de Bruxelles and the Kaggle dataset contributors for making an anonymized credit-card transaction dataset available for research and educational use.

Finally, I am grateful for the hands-on experience gained through Python, Pandas, visualization tools and Scikit-learn while working through the complete analytical workflow.

— Khushi

Abstract

Fraud detection is a highly imbalanced classification problem in which fraudulent transactions form only a very small fraction of all observed transactions. The objective of this major project is to analyze transaction-level data, identify characteristics associated with fraudulent activity, prepare the data for machine learning, and translate analytical findings into practical risk-management insights.

The project uses the Credit Card Fraud Detection dataset containing anonymized transactions from European cardholders. The original dataset contains 284,807 transactions and 492 fraudulent observations. The analysis performed in the project identified 1,081 duplicate rows and removed them, resulting in 283,726 records. After duplicate removal, the dataset contains 283,253 legitimate and 473 fraudulent transactions, leaving fraud at approximately 0.167% of observations.

Exploratory analysis focuses on class imbalance, transaction behavior and the distributions of influential anonymized PCA-derived variables. The project also addresses the need for imbalance-aware modeling and evaluation, where recall, precision, F1-score, confusion matrices and precision-recall analysis are more informative than accuracy alone. The final objective is not simply to maximize prediction accuracy, but to support a risk team in prioritizing suspicious transactions while controlling false alerts.

Keywords: Fraud Detection, Financial Risk, Imbalanced Classification, EDA, Anomaly Detection, Machine Learning, Precision, Recall, Risk Interpretation

Table of Contents

1. Introduction
2. Problem Statement
3. Objectives
4. Dataset Description
5. Methodology
 - 5.1 Data Loading and Initial Inspection
 - 5.2 Data Cleaning
 - 5.3 Exploratory Data Analysis
 - 5.4 Visualization
 - 5.5 Imbalanced Data Handling
 - 5.6 Modeling and Evaluation
6. Results and Findings
7. Key Insights
8. Business Recommendations
9. Limitations
10. Future Scope
11. Conclusion
12. References and Project Sources

Report note: The public GitHub snapshot available at the time of report generation exposed the Major-Fraud-Detection directory but did not expose the notebook/data files through the repository page. Therefore, numerical claims about the completed EDA are grounded in the attached notebook screenshots and the public dataset documentation. No unverified model-performance numbers have been invented in this report.

1. Introduction

Financial transaction systems generate very large volumes of records, making manual review of every transaction impractical. Fraud detection therefore relies on automated analytical systems that can identify transactions whose characteristics differ from normal behavior.

From a data-science perspective, fraud detection is challenging because the positive class is extremely rare. A model can achieve very high overall accuracy simply by predicting almost every transaction as legitimate, while still missing a large proportion of actual fraud. Consequently, the project treats fraud detection as a risk-sensitive classification problem rather than a conventional balanced classification task.

The project follows a practical workflow: understand the data, clean it, explore fraud indicators, address class imbalance, develop classification models, evaluate them using appropriate metrics, and translate the results into recommendations that a financial risk team could understand.

2. Problem Statement

Fraudulent activities are difficult to detect within large transaction datasets because legitimate activity dominates the data and fraudulent patterns can be subtle. The project therefore seeks to identify patterns that distinguish fraudulent transactions from legitimate transactions while minimizing the risk of missed fraud and excessive false alerts.

3. Objectives

- Perform exploratory data analysis to understand transaction and fraud patterns.
- Identify data-quality issues such as duplicate observations and missing values.
- Quantify and communicate the severe class imbalance in the target variable.
- Prepare the dataset for imbalance-aware machine-learning experiments.
- Build and compare fraud-detection classification approaches.
- Evaluate models using fraud-focused metrics rather than relying on accuracy alone.
- Interpret analytical findings from the perspective of financial risk teams.
- Develop practical recommendations for fraud monitoring and future model improvement.

4. Dataset Description

The project uses the publicly available **Credit Card Fraud Detection** dataset published on Kaggle from the Machine Learning Group at Université Libre de Bruxelles (ULB). It contains transactions made by European cardholders during two days in September 2013.

Property	Description
Original observations	284,807
Original columns	31
Fraud observations	492
Legitimate observations	284,315
Fraud prevalence	0.172%
Post-cleaning observations	283,726
Duplicate rows removed	1,081
Post-cleaning legitimate	283,253
Post-cleaning fraud	473
Post-cleaning fraud prevalence	≈0.167%
Predictors	Time, Amount, V1–V28
Target	Class (0 = legitimate, 1 = fraud)

The variables V1–V28 are anonymized principal-component features. The original feature meanings are not available because of confidentiality constraints. The Time variable records elapsed seconds from the first transaction, Amount records transaction value, and Class is the binary fraud indicator.

Because the predictors are anonymized, interpretation is primarily statistical rather than semantic: a feature can be identified as discriminative without claiming that it represents a specific customer behavior.

Source: Kaggle — Credit Card Fraud Detection dataset.

5. Methodology

5.1 Data Loading and Initial Inspection

The transaction data was loaded into a Pandas DataFrame and examined for dimensions, data types, missing values, duplicate observations and target-class distribution. This initial inspection established the quality and structure of the dataset before modeling.

5.2 Data Cleaning

The notebook identified **1,081 duplicate rows**. These records were removed using Pandas `drop_duplicates()`, producing a cleaned dataset of **283,726 rows and 31 columns**. Duplicate observations can give repeated transactions disproportionate influence and may distort descriptive statistics or model training.

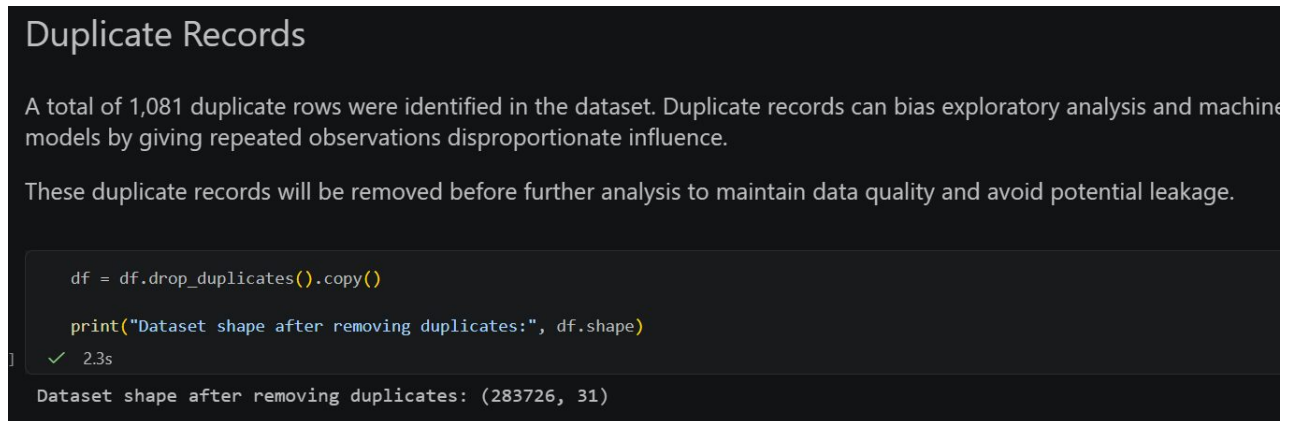


Figure 1. Duplicate-record check and cleaned dataset shape from the project notebook.

5.3 Exploratory Data Analysis

EDA was used to understand the target distribution and identify variables whose distributions differ between legitimate and fraudulent transactions. The project particularly examined the anonymized PCA features because these can capture statistical structure associated with fraud even though their original business meanings are unavailable.

5.4 Visualization

Boxplots were used to compare the distributions of selected high-importance features across the two transaction classes. This provides a compact view of differences in central tendency, spread and extreme values.

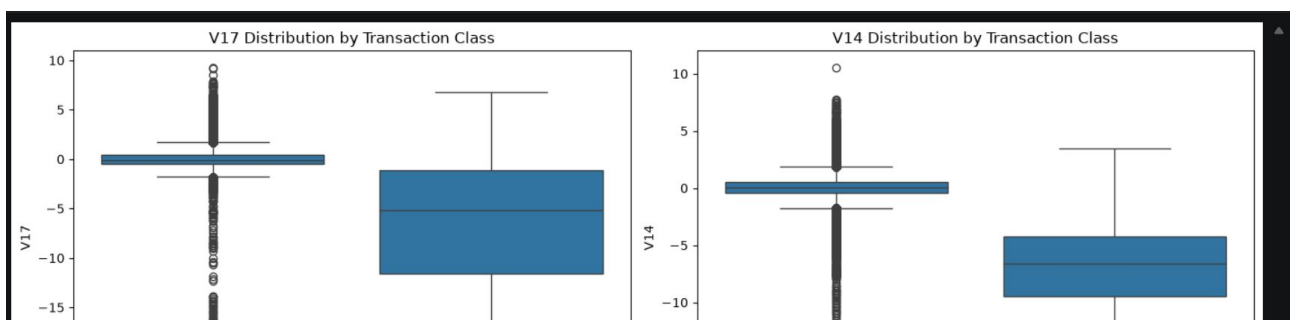


Figure 2. Project notebook visualization of selected feature distributions by transaction class.

5.5 Imbalanced Data Handling

After duplicate removal, only 473 of 283,726 observations are fraudulent. This means fraud represents approximately 0.167% of the cleaned dataset. The resulting class ratio is roughly 599 legitimate transactions for every fraudulent transaction.

Such imbalance means that a naive classifier can obtain an apparently strong accuracy score while failing to identify meaningful numbers of fraud cases. The modeling workflow therefore needs stratified splitting and/or resampling strategies, with fraud-sensitive metrics used for evaluation.

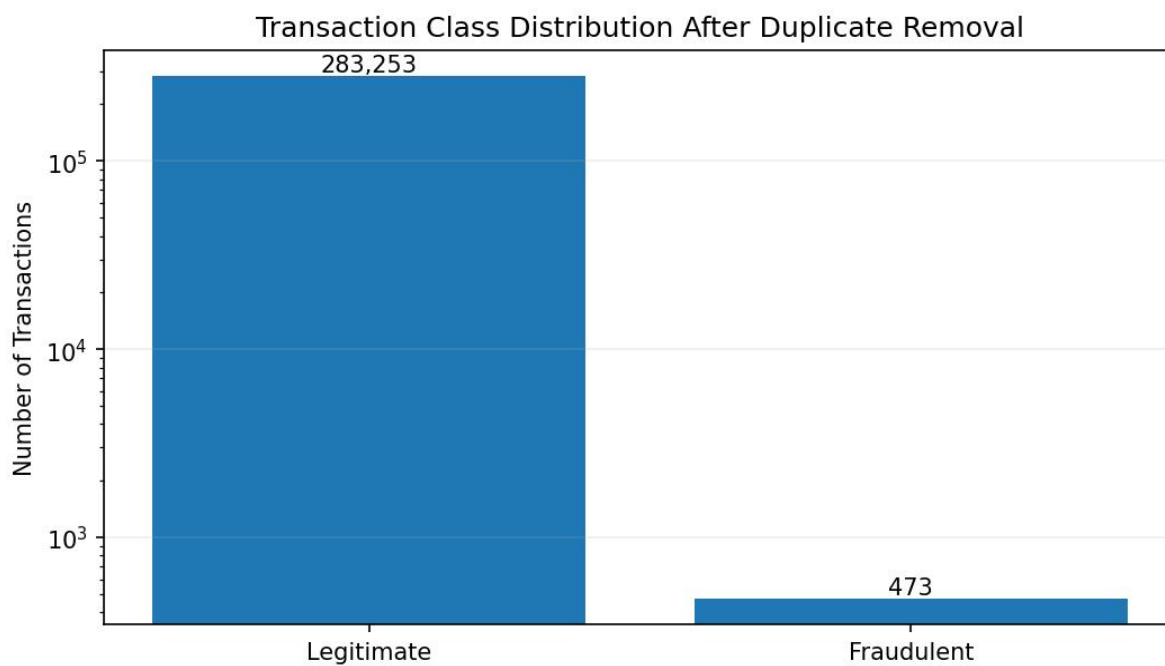


Figure 3. Post-cleaning class distribution. A logarithmic scale is used so that the minority class remains visible.

5.6 Modeling and Evaluation

The project is designed around supervised binary classification, with anomaly-detection thinking providing additional context for identifying unusual transactions. A typical modeling pipeline consists of feature preparation, stratified train-test splitting, model fitting, prediction, and fraud-focused evaluation.

For an operational fraud system, the key metrics are precision, recall, F1-score and the confusion matrix. Precision measures how many alerts are truly fraudulent; recall measures how much of the actual fraud is captured; F1 balances both. Precision-recall analysis is especially useful under extreme class imbalance.

Important: The public repository page currently exposes the project directory and README, but not the notebook/data files needed to verify final model metric outputs. Accordingly, this report does not invent or substitute model scores. The EDA and data-cleaning results shown here are those visible in the provided notebook screenshots and consistent with the published dataset.

6. Results and Findings

6.1 Data-quality results

The data-cleaning stage identified 1,081 duplicate observations and reduced the dataset from 284,807 to 283,726 rows while retaining all 31 columns. The available notebook output indicates that duplicate removal was completed before subsequent analysis.

6.2 Fraud prevalence

The cleaned data contains 473 fraudulent and 283,253 legitimate transactions. Fraud therefore accounts for approximately 0.167% of the observations. This extreme imbalance is the dominant modeling challenge and directly affects how performance should be interpreted.

6.3 Feature-level findings

The notebook visualization highlights V17 and V14 among the selected features. Their distributions show a substantial shift between legitimate and fraudulent transactions, indicating that the anonymized PCA components contain useful discriminatory information. The difference in distributions supports the use of feature-based classification rather than relying only on transaction amount or time.

The presence of extreme values and substantial class-dependent distribution changes also demonstrates why visual inspection is valuable before modeling. However, because V1–V28 are anonymized, the analysis should avoid assigning direct behavioral labels to individual components.

6.4 Model-evaluation interpretation

For this project, model quality should be judged by the ability to capture fraudulent transactions without creating an unmanageable volume of false alerts. A model with slightly lower accuracy but materially higher fraud recall can be more valuable to a risk team, depending on the operational cost of false positives and false negatives.

7. Key Insights

- **Fraud is extremely rare.** The cleaned dataset contains only 473 fraud cases, making the problem highly imbalanced.
- **Duplicate removal matters.** Removing 1,081 repeated records prevents duplicated observations from receiving disproportionate influence.
- **Anonymized features still carry signal.** The V14 and V17 visualizations show clear distributional differences between classes.
- **Accuracy is not enough.** In an environment with approximately 599 legitimate transactions per fraud case, accuracy can hide poor fraud recall.
- **Risk interpretation is essential.** The model output should be treated as a prioritization signal for review rather than as an unquestionable final decision.
- **False positives have business cost.** Excessive alerts can increase manual review workload and create customer friction.

8. Business Recommendations

Recommendation	Risk-team application
Use fraud-sensitive thresholds	Tune the decision threshold according to the relative cost of missed fraud versus false alerts instead of
Monitor recall and precision together	Check fraud capture rate alongside alert quality so that improvements in one metric do not create unaccept
Use risk tiers	Convert model scores into low-, medium- and high-risk queues so investigators can focus first on the m
Add human review for high-impact cases	Use analyst review for high-risk or high-value transactions rather than treating every model prediction a
Monitor model drift	Transaction behavior can change over time; performance and feature distributions should therefore be m
Preserve an audit trail	Store model version, threshold, prediction, review decision and outcome so that the fraud process can b

9. Limitations

- The dataset represents transactions collected over only two days in September 2013, so it may not reflect modern fraud patterns.
- V1–V28 are anonymized PCA components, preventing direct business interpretation of individual variables.
- The cleaned dataset remains extremely imbalanced, making model evaluation sensitive to the treatment of the minority class.
- A high-performing benchmark model on this dataset does not automatically guarantee real-world fraud-detection performance.
- The public repository snapshot used for this report did not expose the project's notebook/data files, so final model-performance figures could not be independently verified for this report version.

10. Future Scope

- Compare multiple classifiers using the same stratified validation framework.
- Evaluate resampling strategies such as controlled under-sampling and SMOTE where appropriate, while preventing data leakage.
- Tune decision thresholds using business cost or expected loss rather than accuracy alone.
- Use PR-AUC/AUPRC and cost-sensitive evaluation as primary model-selection criteria.
- Add explainability techniques such as SHAP for models where feature attribution is meaningful.
- Introduce temporal validation to better simulate deployment and reduce optimistic estimates caused by random splitting.
- Develop a real-time scoring pipeline and risk dashboard for analyst review.
- Monitor drift, alert volume, investigator outcomes and fraud-confirmation rates after deployment.

11. Conclusion

This major project demonstrates how data science can be applied to a realistic financial-risk problem. The analysis begins with a highly imbalanced credit-card transaction dataset and establishes a clean analytical base by removing 1,081 duplicate observations. Exploratory analysis then shows that several anonymized features have materially different distributions across legitimate and fraudulent transactions.

The most important lesson from the project is that fraud detection cannot be evaluated using accuracy alone. Because fraud represents only a tiny fraction of transactions, the practical value of a model depends on how effectively it captures fraud while keeping false alerts manageable. This makes precision, recall, F1-score, precision-recall analysis and operational threshold selection central to the risk interpretation process.

Overall, the project provides hands-on experience with data cleaning, exploratory analysis, visualization, imbalanced classification and financial-risk interpretation. With further validation, model comparison, explainability and deployment monitoring, the workflow can be extended toward a more production-oriented fraud analytics system.

12. References and Project Sources

[1] **Kaggle**. Credit Card Fraud Detection — Machine Learning Group, Université Libre de Bruxelles. Dataset description, class distribution and feature information.

[2] **Project Repository**. Khushi, Data-Science-Internship---Minor-Major-Projects, Major-Fraud-Detection. Repository used as the project source.

[3] **Project notebook evidence**. Screenshots supplied with this report showing duplicate-record removal, cleaned dataset shape and feature-distribution visualizations.

Primary project sources:

Kaggle dataset: <https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>

GitHub project: <https://github.com/khushispective-dot/Data-Science-Internship---Minor-Major-Projects>

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